Yilin Wu

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ACADEMIC BACKGROUND

Carnegie Mellon University, Pittsburgh PA — 2023 - 2027 (expected)
Ph.D. in Robotics, School of Computer Science
Advisor: Andrea Bajcsy

Stanford University, Stanford CA — 2021 - 2023
M.S. in Computer Science, Artifical Inteliigence
Advisor: Dorsa Sadigh

Shanghai Jiao Tong University, Shanghai China — 2016 - 2020
B.S. in Information Security

EXPERIENCE

The Interactive and Trustworthy Robotics Lab (Intent Lab), CMU
Graduate Researcher — 2024.5 - Present
Advised by Andrea Bajcsy
Research on embodied alignment and continual learning for robotic manipulation in the human-centered environment.

Amazon Frontier AI & Robotics Lab, San Francisco, CA
Research Scientist Intern — 2026.5 - 2026.10
Developed Latent Memory For Vision Language Action Models that has flexibility to different VLA Backbones and preserves general capabilities of VLAs with fast inference.

Nvidia Corporation, Seattle, WA
Research Scientist Intern — 2025.5 - 2025.10
Developed methods runtime steering methods for text-action alignment to enable long-horizon manipulation with large reasoning Vision Language Action Models in Seattle Robotics Lab and made a submission to ICRA 2026.

The Robots Perceiving and Doing Lab (R-Pad Lab), CMU
Graduate Researcher — 2023.10 - 2024.4
Advised by David Held
Research on applying reinforcement learning and flow-based methods for long-horizon, contact-rich manipulation, with robust generalization across object geometries, tool morphologies, and diverse tasks.

The Intelligent and Interactive Autonomous Systems Group (ILIAD Lab)
Master Student — 2021.9 - 2023. 6
Advised by Dorsa Sadigh
Research on assistive feeding including bimanual food acquisition and reactive in-mouth bite transfer as well as affordance-based imitation learning for bimanual manipulation.

Nvidia Corporation, Santa Clara, CA
Research Scientist Intern — 2022.6 - 2022.9
Developed methods to improve the efficiency of model-based distributed reinforcement learning in Applied Deep Learning Research Team.

Shanghai Qi Zhi Institute, Shanghai
Research Assistant — 2020.9 - 2021.5
Advised by Yi Wu
Developed methods to improve reinforcement learning algorithms with self-imitation for robotic tasks by automatically discovering subgoals from value function.

Berkeley Artificial Intelligence Research Lab (BAIR Lab), UC Berkeley
Undergraduate Researcher — 2019.1 - 2019.9
Advised by Pieter Abbeel
Proposed a conditional pick-and-place action space and developed a reinforcement learning method with spatial action maps for deformable object manipulation.

PUBLICATIONS

[17] **Yilin Wu**, Rutav Shah, Shuaiyi Huang, Sumedh A. Sontakke, Homanga Bharadhwaj,

Andrea Bajcsy, Max Simchowitz, and Guanya Shi, Memory Without Curation: Learning Multi-Stream Latent Memory for Robot Policies, under review, 2026

[16] **Yilin Wu**, Zilin Si, Zeynep Temel, Oliver Kroemer, Andrea Bajcsy,Inference-time Policy Steering via Vision and Touch, *The Conference on Robot Learning (CoRL)*, 2026

[15] Arnav Kumar Jain [*], **Yilin Wu[*]**, Jesse Farebrother, Gokul Swamy, Andrea Bajcsy, WEAVER, Better, Faster, Longer: An Effective World Model for Robotic Manipulation, under review, 2026

[14] Ran Tian, **Yilin Wu**, Andrea Bajcsy, Good Embodied Reward Models Need Bad Behavior Data, Forty-*Third International Conference on Machine Learning (ICML)*, 2026, **Spotlight top 5%**

[13] Jessie Yuan[*], **Yilin Wu[*]**, Andrea Bajcsy, When to Act, Ask, or Learn: Uncertainty-Aware Policy Steering, *Robotics: Science and Systems (RSS)* , 2026.

[12] **Yilin Wu**, Anqi Li, Tucker Hermans, Fabio Ramos, Andrea Bajcsy[§], Claudia Pérez-D'Arpino[§], Do What You Say: Steering Vision-Language-Action Models via Runtime Reasoning-Action Alignment Verification. *International Conference on Robotics and Automation (ICRA)*, 2026, **Outstanding Paper Award at CVPR 2026 2nd Multimodal Reasoning For Agentic Intelligence Workshop**

[11] **Yilin Wu**, Ran Tian, Gokul Swamy, Andrea Bajcsy, From Foresight to Forethought: VLM-In-the-Loop Policy Steering via Latent Alignment. *Robotics: Science and Systems (RSS)*, 2025, **Best Paper Award at ICLR World Model Workshop**

[10] Ran Tian, **Yilin Wu**, Chenfeng Xu, Masayoshi Tomizuka, Jitendra Malik, Andrea Bajcsy, Maximizing Alignment with Minimal Feedback: Efficiently Learning Rewards for Visuomotor Robot Policy Alignment. Preprint.

[9] Carl Qi[*], **Yilin Wu[*]**, Lifan Yu, Haoyue Liu, Bowen Jiang, Xingyu Lin[†], David Held[†]. Learning Generalizable Tool-use Skills through Trajectory Generation. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[8] Bowen Jiang[*], **Yilin Wu[*]**, Wenxuan Zhou, Chris Paxton, David Held. Spatially-Grounded Motion Primitives for Manipulation. *Robotics: Science and Systems (RSS)*, 2024.

[7] Alexander Khazatsky*, Karl Pertsch*, ... , **Yilin Wu**, ... DROID: A Large-Scale In-the-Wild Robot Manipulation Dataset. *Robotics: Science and Systems (RSS)*, 2024.

[6] Abby O'Neill, ... , **Yilin Wu**, ... Open X-Embodiment: Robotic Learning Datasets and RT-X Models. *International Conference on Robotics and Automation (ICRA), 2024.* **Best Paper Award**

[5] Jennifer Grannen, **Yilin Wu**, Brandon Vu, Dorsa Sadigh. Stabilize to Act: Learning to Coordinate for Bimanual Manipulation. *The Conference on Robot Learning (CoRL)*, 2023, **Oral Presentation**

[4] Lorenzo Shaikewitz[*], **Yilin Wu[*]**, Suneel Belkhale[*], Jennifer Grannen, Priya Sundaresan, Dorsa Sadigh. In-Mouth Robotic Bite Transfer with Visual and Haptic Sensing. *International Conference on Robotics and Automation (ICRA)*, 2023

[3] Jennifer Grannen[*], **Yilin Wu[*]**, Suneel Belkhale, Dorsa Sadigh. Learning Bimanual Scooping Policies for Food Acquisition. *The Conference on Robot Learning (CoRL)*, 2022

[2] Yunfei Li, **Yilin Wu**, Huazhe Xu, Xiaolong Wang, Yi Wu. Solving Compositional Reinforcement Learning Problems via Task Reduction. *The International Conference on Learning Representations (ICLR)*, 2021

[1] **Yilin Wu[*]**, Wilson Yan[*], Thanard Kurutach, Lerrel Pinto, Pieter Abbeel. Learning to Manipulate Deformable Objects without Demonstrations. *Robotics: Science and Systems (RSS)*, 2020

AWARDS

Outstanding Paper Award at CVPR 2026 2nd MMRAgI Workshop	2026
Best Paper Award at ICLR 2025 World Model Workshop	2025
Best Paper Award at ICRA 2024	2024
Graduated with Honor (Outstanding Graduate of Shanghai)	2020
Hongyi Scholarship	2019
National Scholarship	2017

ACADEMIC SERVICE

External Reviewer for Conferences, Journals

- IEEE Robotics and Automation Letters (RAL): 2024
- International Symposium of Robotics Research (ISRR): 2024
- Conference on Robot Learning (CoRL): 2024, 2025, 2026
- International Conference on Intelligent Robots and Systems (IROS): 2023, 2024
- Robotics: Science and Systems (RSS): 2024, 2025
- International Conference on Robotics and Automation (ICRA): 2024, 2025, 2026
- International Conference on Learning Representations (ICLR): 2024, 2025

Teaching Assistant for Computer Science Courses

- Carnegie Mellon University 16-867: Human-Robot Interaction (Fall 2025)
- Carnegie Mellon University 16-831: Introduction to Roobt Learning (Fall 2024)
- Stanford CS 221: Artificial Intelligence: Principles and Techniques (Spring 2023, Spring 2022, Fall 2021)
- Stanford CS 148: Introduction to Computer Graphics and Imaging (Fall 2022)
- Stanford CS182: Ethics, Public Policy, and Technological Change (Winter 2023, Winter 2022)